FN5 Worked Examples: Combining and Composing Functions

Recording companion for the FN5 CheckIt support video — Math 172, Santa Ana College

How to use this document

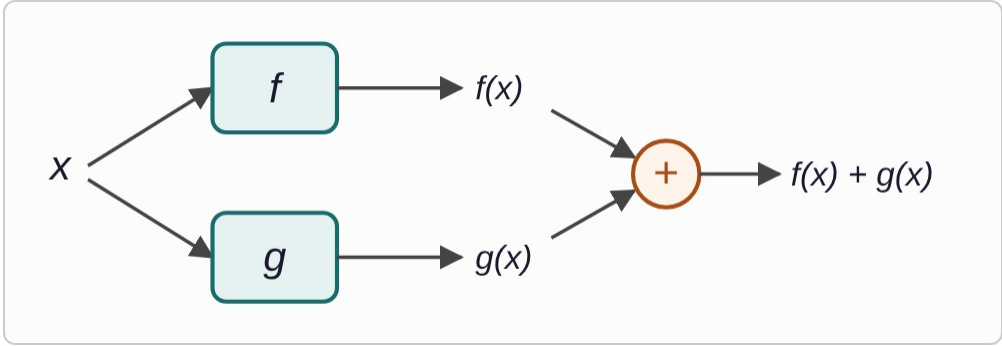
- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

Am I combining two outputs, or feeding one function's output into the other? For $f + g$, $f - g$, $f \cdot g$ and $f \div g$, both functions take the same input, and you combine the two outputs, keeping each function in parentheses. For $f(g(x))$, the inside function goes first: replace every x in f with the whole expression $g(x)$, in parentheses. For a number, work from the inside out: find $g(n)$, then put that number into f.

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Example 1: Sum



Plotted data: A diagram. The same input, x, goes into two machines, f and g. Their outputs, f(x) and g(x), meet at a plus sign, giving f(x) + g(x).

$$f(x)=3x-4$$
$$g(x)=2x^2+5$$

Question

Find f(x) + g(x).

- **A.** $2x^2+3x+1$
- **B.** $5x^2+1$
- **C.** $2x^2+3x+9$
- **D.** $2x^2+3x-1$

Answer and reasoning

Add the two formulas and combine like terms: $(3x-4)+(2x^2+5)=2x^2+3x+1$. The x^2 term and the x term stay separate, so $5x^2+1$ combines terms that are not alike. $+9$ treats the -4 as $+4$, and -1 works out $4-5$ instead of $-4+5$. ***$2x^2+3x+1$.***

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Example 2: Difference

$$f(x)=3x-4$$
$$g(x)=2x^2+5$$

Question

Find f(x) − g(x).

- **A.** $-2x^2+3x+1$
- **B.** $2x^2-3x+9$
- **C.** $-2x^2+3x-9$
- **D.** $-2x^2-3x-9$

Answer and reasoning

Put all of $g(x)$ in parentheses and subtract each of its terms: $(3x-4)-(2x^2+5)=3x-4-2x^2-5=-2x^2+3x-9$. $+1$ subtracts the $2x^2$ but adds the 5; $2x^2-3x+9$ is $g(x)-f(x)$, the other order; and $-3x$ changes the sign of a term that belongs to f. **$-2x^2+3x-9$.**

Example 3: Product

$$f(x)=3x-4$$
$$g(x)=2x^2+5$$

Question

Find f(x) · g(x), multiplied out.

- **A.** $6x^3-20$
- **B.** $6x^3-8x^2+15x-20$
- **C.** $6x^3+8x^2+15x-20$
- **D.** $5x^3-8x^2+15x-20$

Answer and reasoning

Multiply each term of $3x-4$ by each term of $2x^2+5$: $3x\cdot 2x^2=6x^3$, $3x\cdot 5=15x$, $-4\cdot 2x^2=-8x^2$, and $-4\cdot 5=-20$. Together: $6x^3-8x^2+15x-20$. $6x^3-20$ keeps only the first and last products; $+8x^2$ loses the minus on the 4; and $5x^3$ adds 3 and 2 instead of multiplying them. CheckIt may show the product unexpanded, as $(3x-4)(2x^2+5)$; that is the same function. ***$6x^3-8x^2+15x-20$*.**

Example 4: Quotient

$$f(x)=3x-4$$
$$g(x)=2x^2+5$$

Question

Find f(x) divided by g(x).

A. $(3x-4)/(2x^2+5)$
B. $(2x^2+5)/(3x-4)$
C. $3x-4/(2x^2+5)$
D. $3/2x-4/5$

Answer and reasoning

The whole formula for *f* goes on top and the whole formula for *g* goes on the bottom: $(3x-4)/(2x^2+5)$. Nothing cancels, because no factor is shared. Flipping it gives *g(x)* divided by *f(x)*; $3x-4/(2x^2+5)$ divides only the -4, which is what happens when the parentheses around $3x-4$ are left off; and dividing term by term, $3/2x-4/5$, is not allowed when the bottom has more than one term. ***(3x−4)/(2x²+5).***

Example 5: Compose 1

$$f(x)=2x+1$$
$$g(x)=x^2-3$$

Question

Find f(g(x)).

- **A.** $4x^2+4x-2$
- **B.** $2x^2-2$
- **C.** $2x^3+x^2-6x-3$
- **D.** $2x^2-5$

Answer and reasoning

g is inside, so its whole formula replaces the x in f: $f(g(x))=2(x^2-3)+1=2x^2-6+1=2x^2-5$. $4x^2+4x-2$ is $g(f(x))=(2x+1)^2-3$, the other order; $2x^2-2$ leaves out the parentheses, so the 2 never reaches the -3; and $2x^3+x^2-6x-3$ is the product $f(x)\cdot g(x)$, not a composition. ***$2x^2-5$*.**

Example 6: Compose 2

$$h(x)=\sqrt{(2x+3)}$$
$$k(x)=x^2+1$$

Question

Find h(k(x)).

A. $2x+4$
B. $\sqrt{(2x^2+5)}$
C. $\sqrt{(2x^2+4)}$
D. $\sqrt{(2x^2+3)}+1$

Answer and reasoning

k is inside, so x^2+1 replaces the x under the root: $h(k(x))=\sqrt{(2(x^2+1)+3)}=\sqrt{(2x^2+2+3)}=\sqrt{(2x^2+5)}$. $2x+4$ is $k(h(x))=(\sqrt{(2x+3)})^2+1$, the other order; $\sqrt{(2x^2+4)}$ multiplies only the x^2 by 2, not the 1; and $\sqrt{(2x^2+3)}+1$ puts the $+1$ from k outside the root instead of inside, where x was. **$\sqrt{(2x^2+5)}$.**

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Example 7: Compose 3

$$r(x)=2x/(x+3)$$
$$q(x)=4x-1$$

Question

Find r(q(x)).

A. $8x/(x+3)-1$
B. $(8x-2)/(x+3)$
C. $(8x-2)/(4x+2)$
D. $(8x-1)/(4x+2)$

Answer and reasoning

Replace both x's in r, top and bottom, with $(4x-1)$: $r(q(x))=(2(4x-1))/((4x-1)+3)=(8x-2)/(4x+2)$. It also simplifies to $(4x-1)/(2x+1)$, dividing top and bottom by 2. $8x/(x+3)-1$ is $q(r(x))$, the other order; $(8x-2)/(x+3)$ replaces only the x on top; and $8x-1$ multiplies only the $4x$ by 2. ***(8x−2)/(4x+2).***

Example 8: Which One?

$$h(x)=x^3$$
$$p(x)=x+2$$

Question

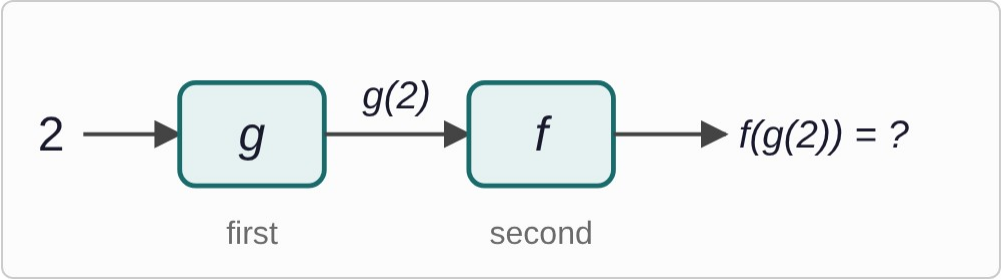
Which formula is h(p(x))?

- **A.** x^3+2
- **B.** x^4+2x^3
- **C.** x^3+x+2
- **D.** $(x+2)^3$

Answer and reasoning

p is inside, so its output, $x+2$, is what gets cubed: $h(p(x))=(x+2)^3$. x^3+2 cubes first and then adds 2, which is $p(h(x))$; x^4+2x^3 is the product $h(x)\cdot p(x)$; and x^3+x+2 is the sum $h(x)+p(x)$. ***$(x+2)^3$*.**

Example 9: Evaluate 1



Plotted data: A diagram of two machines in a row. The input 2 goes into machine g first. Its output, g(2), goes into machine f second. The final output, f(g(2)), is marked with a question mark.

$$f(x)=3x-2$$
$$g(x)=x^2+1$$

Question

Find f(g(2)).

- **A.** *17*
- **B.** *13*
- **C.** *20*
- **D.** *5*

Answer and reasoning

Inside first: $g(2)=2^2+1=5$. Then put 5 into f: $f(5)=3(5)-2=13$. 17 is $g(f(2))=g(4)$, the other order; 20 is the product $f(2)\cdot g(2)=4\cdot 5$; and 5 stops halfway, at $g(2)$. ***f(g(2))=13*.**

Example 10: Evaluate 2

$$p(x)=x^2-4$$
$$q(x)=\sqrt{(2x+1)}$$

Question

Find q(p(3)).

- **A.** $\sqrt{11}$
- **B.** 3
- **C.** $\sqrt{7}$
- **D.** $\sqrt{10}+1$

Answer and reasoning

Inside first: $p(3)=9-4=5$. Then $q(5)=\sqrt{(2(5)+1)}=\sqrt{11}$, an exact answer to leave as it is. 3 is $p(q(3))=(\sqrt{7})^2-4$, the other order; $\sqrt{7}$ is $q(3)$, which skips p; and $\sqrt{10}+1$ takes the root of $2(5)$ before adding the 1. ***q(p(3))=√11*.**

Example 11: Evaluate 3

$$r(x)=3x/(x+2)$$
$$s(x)=2x-5$$

Question

Find r(s(4)).

A. -1
B. 2
C. $3/2$
D. $9/5$

Answer and reasoning

Inside first: $s(4)=2(4)-5=3$. Then 3 replaces both x's in r: $r(3)=(3(3))/(3+2)=9/5$. -1 is $s(r(4))=s(2)$, the other order; 2 is $r(4)=12/6$, which skips s; and $3/2=9/6$ puts the 3 on top but leaves the original 4 on the bottom. ***r(s(4))=9/5*.**

Example 12: The Other Order

$$h(x)=x^2+2$$
$$k(x)=3x-1$$
$$h(k(x))=9x^2-6x+3$$

Question

Like CheckIt, the formula for h(k(x)) is given first. Now find k(h(−1)).

A. *18*
B. *8*
C. *3*
D. *2*

Answer and reasoning

The question asks for $k(h(-1))$, the opposite order from the formula, so that formula does not help. Inside first: $h(-1)=(-1)^2+2=3$. Then $k(3)=3(3)-1=8$. 18 puts -1 into the $h(k(x))$ formula, the wrong order; 3 stops at $h(-1)$; and 2 squares -1 as -1, so $h(-1)$ comes out 1 and $k(1)=2$. ***k(h(−1))=8*.**

Example 13: Slip 1

$$f(x)-g(x)=(2x^2+4x)-(2x-3)$$
$$=2x^2+4x-2x-3$$
$$=2x^2+2x-3$$

Question

A student wrote this work. What is the mistake, if any?

A. There is no mistake.
B. It should be g(x) − f(x), the other way around instead.
C. The minus must reach the −3 too: 2x² + 2x + 3.
D. The x terms should add instead, giving 2x² + 6x − 3.

Answer and reasoning

Subtracting $(2x-3)$ means subtracting both terms: $-(2x-3)=-2x+3$. The student changed the sign of $2x$ but not of -3. Correct: $2x^2+2x+3$. The question asked for $f(x)-g(x)$, so the order is right; and $4x-2x$ really is $2x$, not $6x$. ***f(x)−g(x)=2x²+2x+3*.**

Example 14: Slip 2

$$f(x)\cdot g(x)=(2x+3)(3x-1)=6x^2-3$$

Question

A student wrote this work. What is the mistake, if any?

A. There is no mistake.
B. (3)(−1) is +3, so it should be $6x^2 + 3$.
C. A product means composition, so it is $2(3x - 1) + 3$.
D. The middle terms were dropped: $6x^2 + 7x - 3$.

Answer and reasoning

Every term multiplies every term: $2x\cdot 3x=6x^2$, $2x\cdot(-1)=-2x$, $3\cdot 3x=9x$, and $3\cdot(-1)=-3$. The middle terms combine to $7x$: $6x^2+7x-3$. $(3)(-1)$ is -3, which the student had right; and $2(3x-1)+3$ is $f(g(x))$, a composition, not a product. ***$6x^2+7x-3$.***

Example 15: Slip 3

$$f(x)=x^2, \quad g(x)=x+5$$

Question

A student says f(g(x)) = x² + 5. What is the mistake, if any?

A. There is no mistake.
B. That is g(f(x)); f(g(x)) is (x + 5)².
C. Composition multiplies them: x²(x + 5).
D. Square each of the two terms: x² + 25.

Answer and reasoning

In *f(g(x))*, *g* goes first and *f* squares its output: $f(g(x))=(x+5)^2=x^2+10x+25$. The student squared first and then added 5, which is $g(f(x))$. $x^2(x+5)$ is the product $f(x)\cdot g(x)$; and $(x+5)^2$ is not x^2+25: squaring a sum has a middle term, $10x$. ***$f(g(x))=(x+5)^2=x^2+10x+25$.***

Example 16: Slip 4

$$f(x)=\sqrt{(2x+1)},\quad g(x)=x^2-1$$
$$f(g(2))=f(3)=\sqrt{7}$$

Question

A student wrote this work. What is the mistake, if any?

A. There is no mistake.
B. Apply f first: g(f(2)) = g(√5) = 4.
C. It should be √3, since g(2) = 3.
D. It should be 7, since the root and the square cancel.

Answer and reasoning

Inside first: $g(2)=4-1=3$. Then $f(3)=\sqrt{(2(3)+1)}=\sqrt{7}$. The student is right. $g(f(2))$ is the other order; $\sqrt{3}$ puts 3 under the root without doubling it and adding 1; and nothing cancels, because the root belongs to f and the square belongs to g. **There is no mistake: $f(g(2))=\sqrt{7}$.**

Example 17: Slip 5

$$(f(x))/(g(x))=(2x^2+3)/2x=x+3$$

Question

A student simplified this quotient by canceling 2x. What is the mistake, if any?

A. There is no mistake.
B. Flip it: the quotient should be 2x over (2x² + 3).
C. 2x is not a factor of 2x² + 3, so it cannot cancel.
D. It should be x + 3/2, dividing just the 3 by the 2.

Answer and reasoning

Only a factor of the whole top can cancel, and $2x$ divides the $2x^2$ but not the 3. Splitting the fraction gives $(2x^2)/2x+3/2x=x+3/2x$, or leave it as $(2x^2+3)/2x$, the way CheckIt does. The order f over g is right as written; and $x+3/2$ forgets that the 3 is divided by $2x$, not by 2. ***(2x²+3)/2x=x+3/2x*, not *x+3*.**